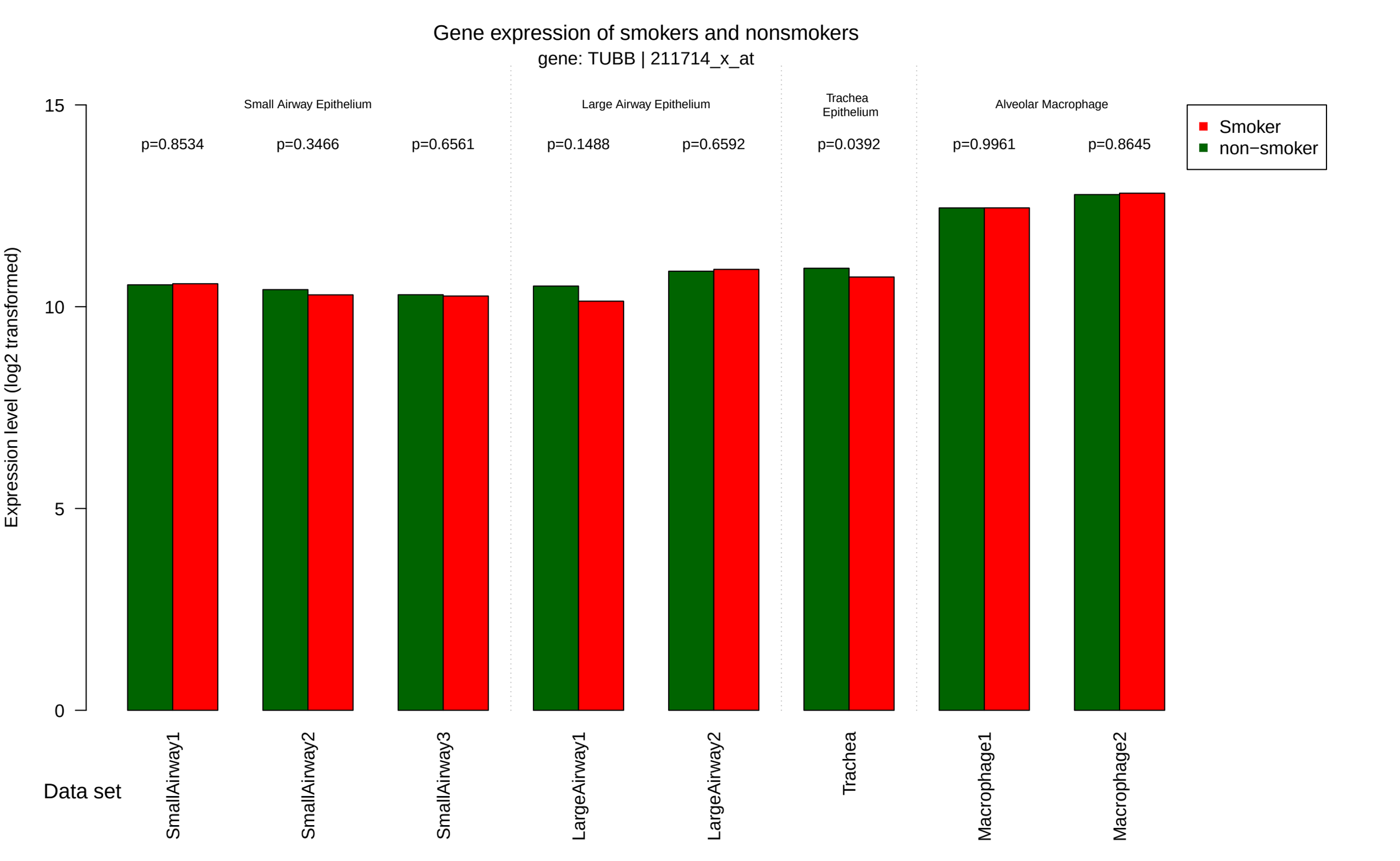
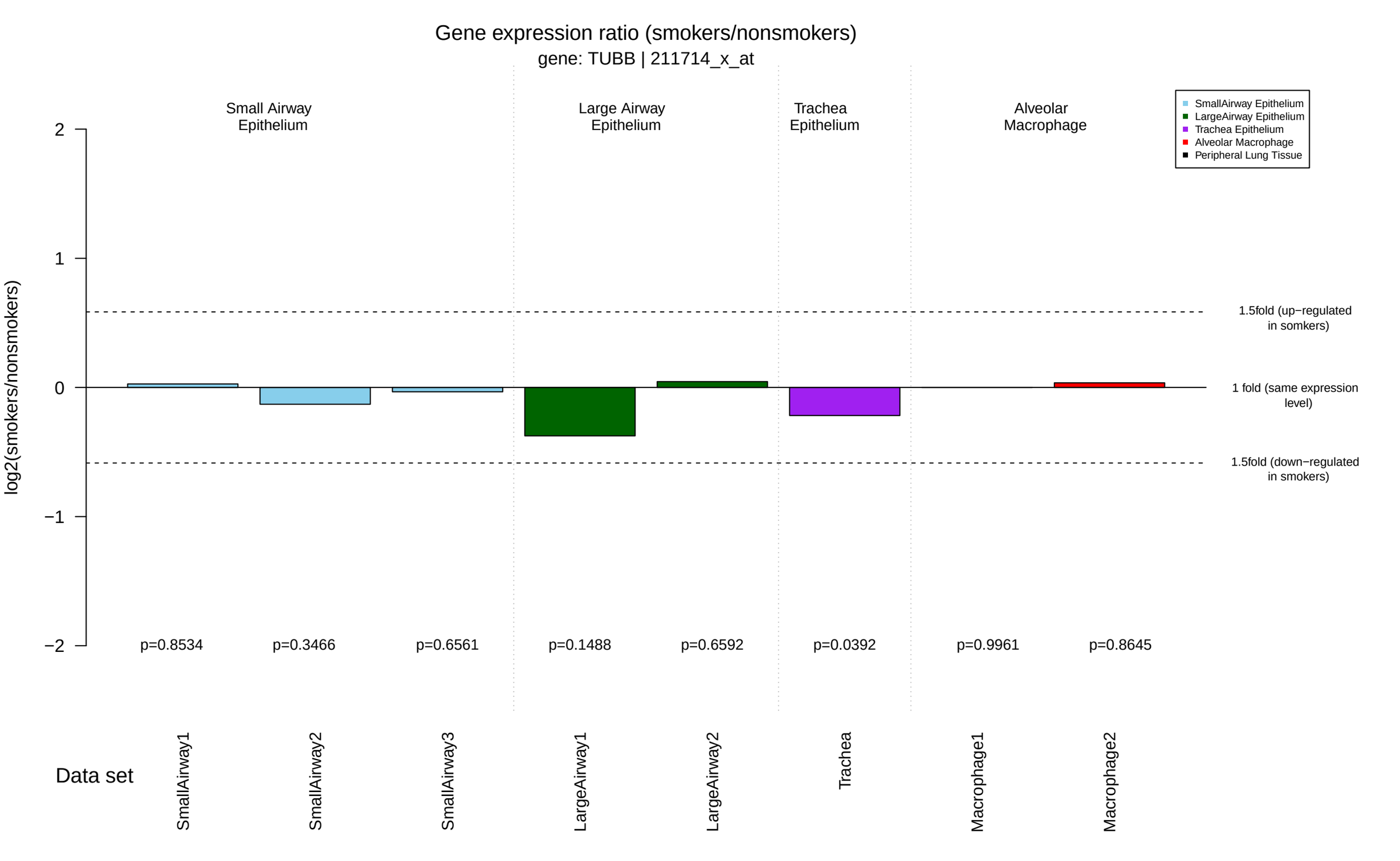
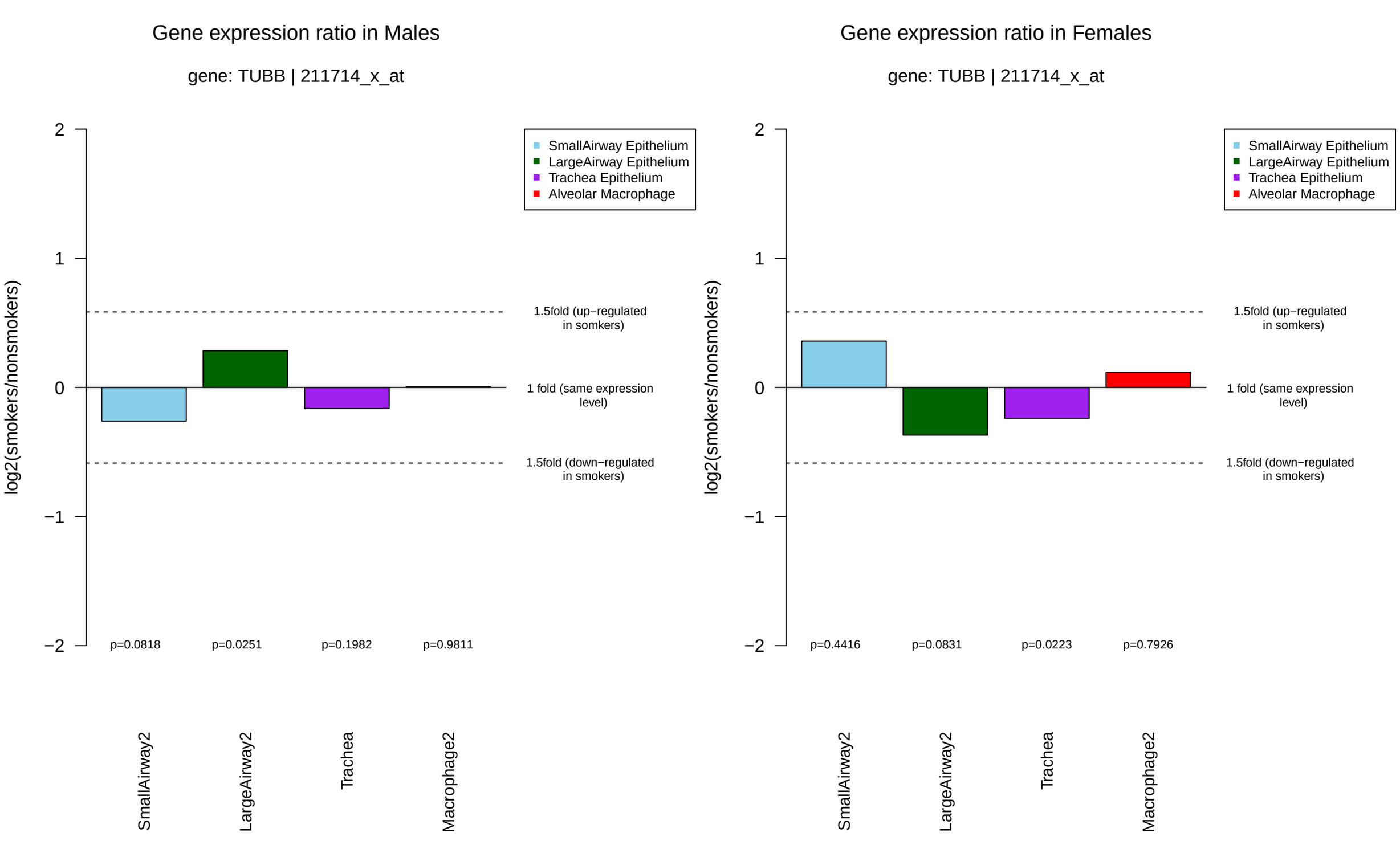
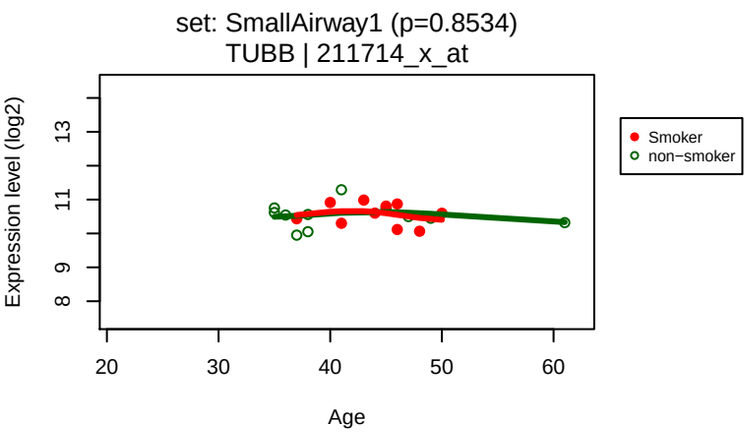
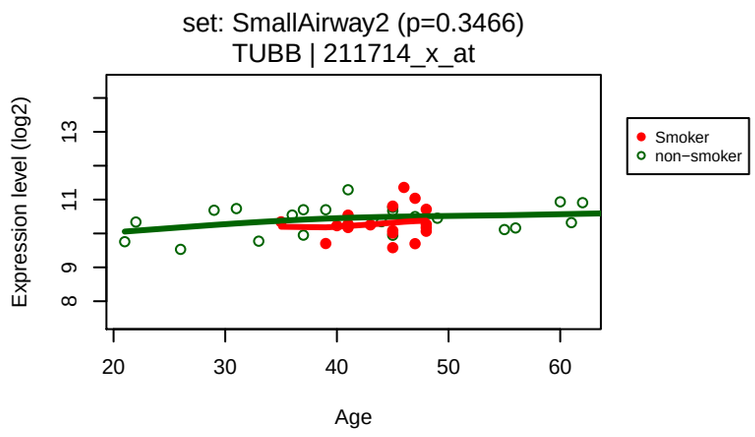
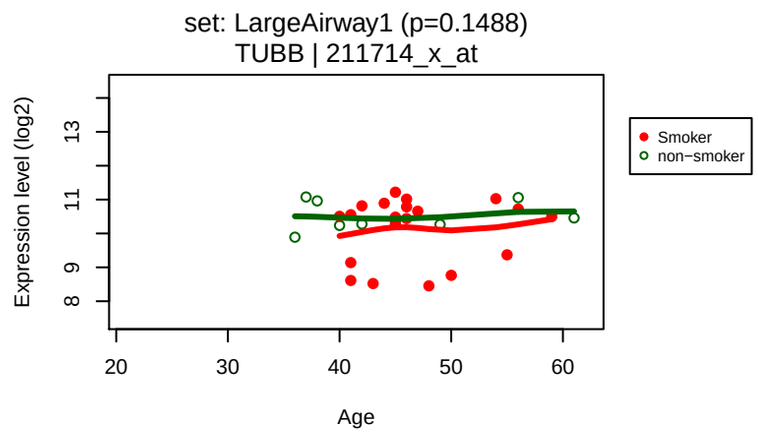

Gene expression of smokers and nonsmokers
gene: TUBB | 211714_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8534
p=0.3466
p=0.6561
p=0.1488
p=0.6592
p=0.0392
p=0.9961
p=0.8645
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

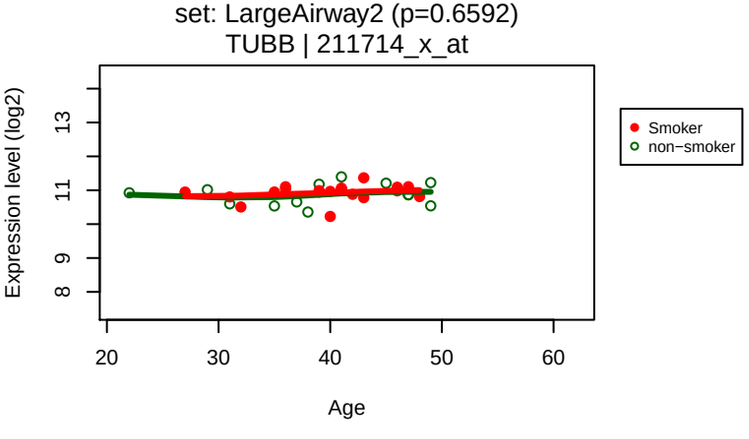

Gene expression ratio (smokers/nonsmokers)
gene: TUBB | 211714_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8534
p=0.3466
p=0.6561
p=0.1488
p=0.6592
p=0.0392
p=0.9961
p=0.8645
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

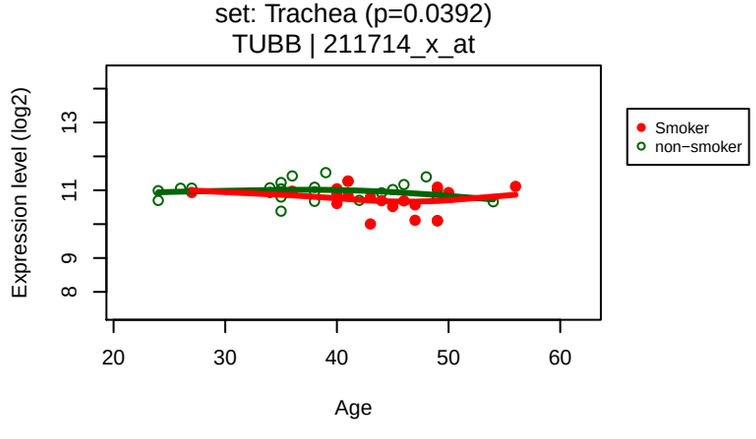

Gene expression ratio in Males
gene: TUBB | 211714_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0818
p=0.0251
p=0.1982
p=0.9811
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TUBB | 211714_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4416
p=0.0831
p=0.0223
p=0.7926
SmallAirway2
LargeAirway2
Trachea
Macrophage2

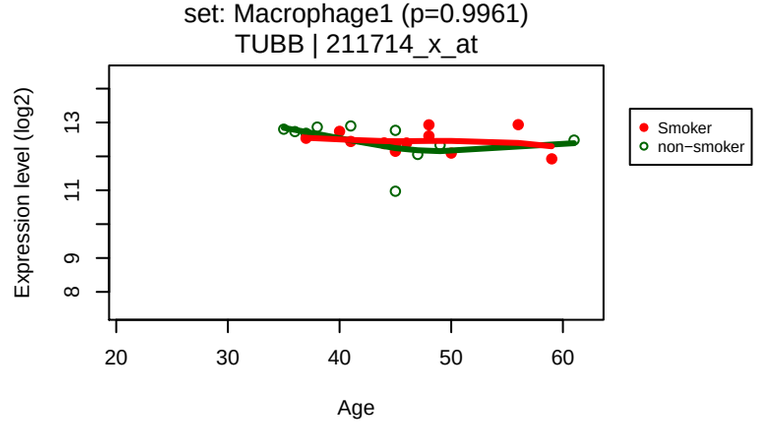

set: SmallAirway1 (p=0.8534)
TUBB | 211714_x_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker

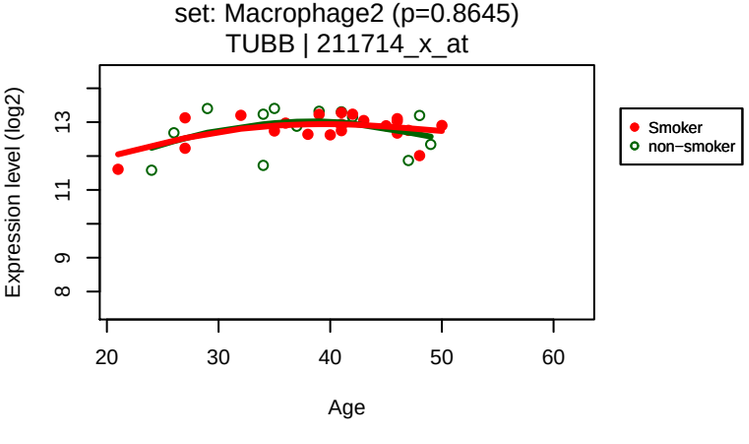

set: SmallAirway2 (p=0.3466)
TUBB | 211714_x_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker

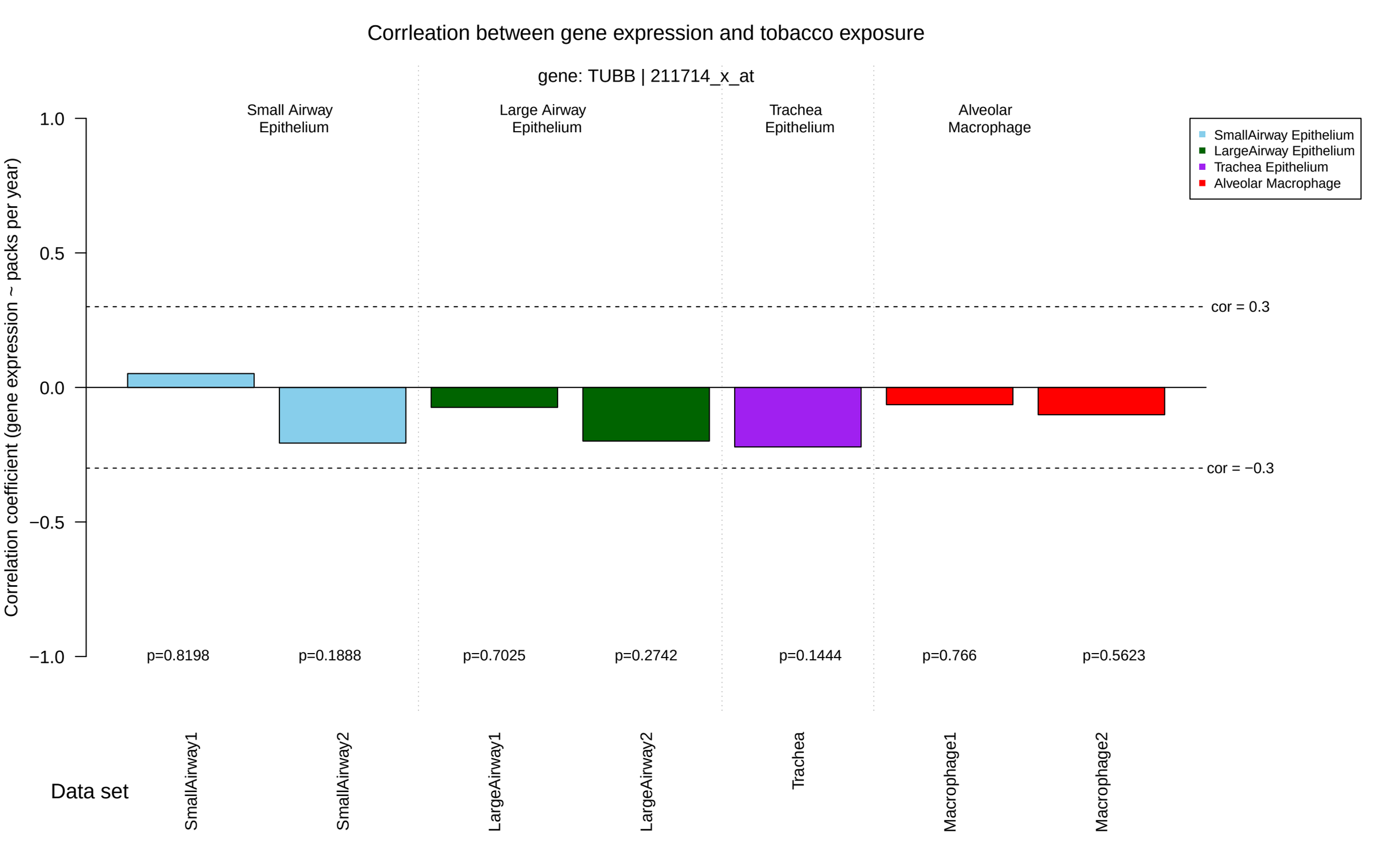

set: LargeAirway1 (p=0.1488)
TUBB | 211714_x_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6592)
TUBB | 211714_x_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0392)
TUBB | 211714_x_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9961)
TUBB | 211714_x_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8645)
TUBB | 211714_x_at
Expression level (log2)
8
9
11
13
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TUBB | 211714_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8198
p=0.1888
p=0.7025
p=0.2742
p=0.1444
p=0.766
p=0.5623
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2